

Music-Gen v4 — Cycles 57-59

2026-09-05

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Music-Gen v4 — Cycles 57-59

Abstract

Cycles 57-59 mark the campaign’s exit from the preservation-only heartbeat cadence and the first substantive post-c24 non-Chicken-Grease bass advance. Two operator directives arrived in `live_guidance` during the range and were correctly acted upon: an operator 2026-09-04 distance-semantics ruling that formally set the 0.40 embedding-cosine value as a degenerate-floor upper bound under distance semantics (resolving what had been the metric-semantics escalation open since c16); and an operator 2026-09-05 omnibus that lifted `SF2_CONFIRMED` campaign-wide under an OPT1-extension best-of-search-across-families acceptance policy (resolving what had been the non-CG bass acceptance-policy escalation open since c7), formally retired the preservation-spin cadence (point 4), and enumerated the remaining M-V4-PROFILES closure work as a seven-item punch list (5(a)-(g): four more bass/drums cells + guitar/piano/other/vocals stems + A/B deliveries + generator batch + completion report v3). Cycle 57 discharged the initial replanning under the arrived directives. Cycle 58 attempted the first stage-2 fine fit under the lifted acceptance policy but hit a P0.1 disk-clearance blocker (workspace above the c27 df-guard comfortable margin) and correctly established the `mv → tools/stale/ + explicit _prune_stale_sweep_audio` invocation pattern from the c27 hygiene module as the workaround when `rm -rf` was refused; the cycle also landed two new tests (`test_43`, `test_44`) on the c30 legacy-mode regression suite and left a small carryover (I-2, `test_12` SHA drift pending mechanical fix). Cycle 59 landed the first substantive advance: `data/v4/profiles/51e433ade2a845e1/bass.json` (SHA `a4d4b47ebd9edd0e...`) is the Rome bass profile v1 selected via the extended `fine_fit_sf2_v2.py` stage-2 driver under the OPT1-extension acceptance policy, with top-1 candidate at bank 0 program 4, composite score 323.55, and `emb_cos_vggish` distance 0.1677 — well under the 0.40 degenerate-floor upper bound per the operator distance-semantics ruling. `bass.replay_proof.json` returned `REPLAY_PROOF_HOLDS` with replay SHA `594f7d4b...` (byte-determinism ×2). The verdict field carries a new (unfrozen) enum label `SF2_CONFIRMED_provisional` to encode “OPT1-lifted acceptance pending sibling-cell replication” — honest labeling that surfaces the promotion-criterion question for near-term formalization. The `supersedes_path` field is a string pointing at the c24 `SF2_RULED_OUT` verdict on Rome bass per the c14 lemma. The OP-1 serial-lock engaged on the `fine_fit_sf2_v2.py` invoca-

tion (data/v4/_run/fine_fit_serial_lock acquired and released cleanly). A single milestone-level ledger event under M-V4-PROFILES was emitted honoring the v4-relaxed cadence — no per-cycle preservation sidecar was emitted per the operator’s point-4 directive. None of the six previously-standing operator escalation memos were resurrected; all six are now formally closed on the substantive side (composite-FP-drift, non-CG bass acceptance policy, metric-semantics, drums-fine, v2-bass-fine, guitar-fine) either by the operator omnibus lifting the substantive constraint or by the distance-semantics ruling formalizing the metric that had been contested. env_pin_sha256=2ac444c3...a922ca canonical 7-key unchanged at pre-launch; FD-16(a) re-issue not triggered; M-V4-CERT-1 E2E_DETERMINISM_HOLDS pre-launch and the certificate remained valid across the fine-fit. Independent audit at range close returned **VALIDATED with observations**: three low-severity observations queued for the next cycle (O-1 SF2_CONFIRMED provisional enum-extension formalization; O-2 sweep-hygiene per-candidate-delete contract clarity in fine_fit_sf2_v2.py; O-3 workspace disk margin thin at 85%-used / 5.9G-free at range close, below the working margin for the next-cycle punch list). Zero CRITICAL, zero HIGH, zero MODERATE. Rome bass profile is the first M-V4-PROFILES-1 cell filled since the range that closed CG at c24; four bass/drums cells plus the remaining punch-list items are queued for the next range’s execution.

Introduction

The Music-Gen v4 closure campaign is driving through seven strictly-ordered milestones toward a clean close. Prior ranges had brought the campaign into a preservation-only heartbeat cadence with all six operator-authority escalations preserved blocked_on_operator=true across fourteen consecutive substantive-heartbeat cycles under the c36 auditor’s terminal contracts. The cadence maintained the campaign in a state from which any incoming operator decision could trigger substantive execution in a single downstream cycle. Cycles 57-59 are the first range in which such decisions arrived.

Two operator directives landed in live_guidance during the range: an operator 2026-09-04 distance-semantics ruling that formally fixed the semantics of the embedding_cos_vggish field as distance (with 0.40 as a degenerate-floor upper bound rather than any similarity threshold), and an operator 2026-09-05 omnibus that carried multiple decisions in one message — lifting SF2_CONFIRMED campaign-wide under an OPT1-extension best-of-search-across-families acceptance policy (point 3), formally retiring the preservation-spin cadence (point 4), and enumerating a seven-item punch list for the remainder of M-V4-PROFILES-1 (points 5(a)-(g)). The range’s task was to act on both directives without re-litigating the resolved questions, without carrying forward preservation-only work that the operator had explicitly ended, and without over-reaching beyond the scope the punch list actually authorized.

Approach

Cycle 57 (directive arrival and replanning). Detected both operator directives in live_guidance. Formally recorded the closure of the six previously-standing operator escalation memos on the substantive side: the composite-FP-drift, non-CG bass acceptance policy, and metric-semantics escalations are resolved by the two directives; the three per-driver fine-fit HALT escalations (drums, v2-bass, guitar) are subsumed into the composite-FP-drift resolution via the c32 consolidation memo. Re-scoped remaining work per the operator’s seven-item punch list. Exited the preservation-only cadence per the operator’s point-4 directive — no per-cycle preservation sidecars emitted from this cycle onward.

Cycle 58 (first stage-2 attempt; disk-clearance blocker; test extensions). Attempted the first stage-2 fine fit under the lifted acceptance policy but hit a P0.1 disk-clearance blocker at the workspace-margin check. Established the workaround pattern when rm -rf was refused:

mv accumulated sweep audio to tools/stale/ and explicitly invoke `_prune_stale_sweep_audio` from the c27 hygiene module. Landed two new tests (test_43, test_44) on tests/test_c30_legacy_mode_regression.py extending the c30 legacy-mode regression suite. Carried forward a small residual: I-2 test_12 SHA drift, a mechanical pin update pending non-blocking for c48 landing but slotted for c49's queue.

Cycle 59 (Rome bass profile v1 landed). Executed the stage-2 fine fit for Rome bass under the extended `fine_fit_sf2_v2.py` driver with OP-1 SerialLock engaged (data/v4/_run/fine_fit_serial_lock acquired and released cleanly). The 217-cell stage-2 sweep produced a leaderboard from which the top-1 candidate was selected: bank 0, program 4, composite score 323.55, `emb_cos_vggish` distance 0.1677 — well under the 0.40 degenerate-floor upper bound per the operator distance-semantics ruling. Emitted data/v4/profiles/51e433ade2a845e1/bass.json (SHA a4d4b47ebd9edd0e...) as the pinned profile with the verdict field carrying a new (unfrozen) enum label `SF2_CONFIRMED_provisional` — honest labeling of the “OPT1-lifted acceptance pending sibling-cell replication” semantic that the operator omnibus opened without pre-declaring the promotion criterion. `supersedes_path` field is a string pointing at the c24 `SF2_RULED_OUT` verdict on Rome bass per the c14 lemma. Emitted `bass.replay_proof.json` with `REPLAY_PROOF_HOLDS` and replay SHA 594f7d4b... under byte-determinism $\times 2$ per FD-16(c). Emitted a single milestone-level ledger event under M-V4-PROFILES honoring the v4-relaxed cadence — no per-cycle preservation sidecar. Did not re-run c30 legacy-mode regression tests (no code touched in the emitter-writer-boundary); the c34 OPT_B exemption chain remains intact with `long_exposure/` still ABSENT and docs/emitter_exemption_policy.md byte-identical.

Discipline guards asserted across the range. All AST-scannable invariants pass: no PRNG imports, no `sidecar_nonfactor` cross-contamination, no VST3 state APIs, no `--verify-det bypass`, /usr/bin/python3 interpreter guard on the fine-fit invocation. `supersedes_path` typed as string per c14 lemma on the Rome bass profile (never list). OP-1 serial-lock engaged and released cleanly. Canonical 7-key `env_pin_sha256=2ac444c3...a922ca` at pre-launch unchanged; FD-16(a) certificate re-issue not triggered. FD-16(c) replay-proof $\times 2$ per render family per song satisfied. No wait-on-operator memo emitted (banned per operator 2026-09-03 point 2). None of the six previously-standing operator escalation memos resurrected; all six now formally closed on the substantive side. FD-1 honored: no tuning, no retry, no fallback observed in the stage-2 fine fit — the top-1 candidate was pinned as the deterministic output of the driver under the specified environment pin.

Findings

Two operator directives arrived and were correctly acted upon

The range is the first in which operator adjudications actually landed in `live_guidance`. Both were acted upon precisely as their text specified:

- **Operator 2026-09-04 distance-semantics ruling.** Formalized `embedding_cos_vggish` as distance (not similarity), with the retained 0.40 value as a degenerate-floor upper bound (candidates above 0.40 are degenerate; candidates below are eligible for acceptance under other policy). This resolves the metric-semantics escalation open since c16.
- **Operator 2026-09-05 omnibus.** Point 3: OPT1 extension — `SF2_CONFIRMED` lifted campaign-wide under best-of-search across families as the acceptance policy. Point 4: preservation-spin cadence formally retired; no per-cycle preservation sidecars from this point. Points 5(a)-(g): remaining M-V4-PROFILES-1 punch list — four more bass/drums cells + guitar/piano/other/vocals stems + A/B deliveries + generator batch + completion report v3.

The range acted on both. The composite-FP-drift, non-CG bass acceptance-policy, and metric-semantics escalations closed on the substantive side; the three per-driver fine-fit HALT escalations subsume into the composite-FP-drift resolution via the c32 consolidation memo. The preservation-only cadence exited at Cycle 57. The remaining work is scoped per the operator punch list.

Rome bass profile v1 is the first substantive post-c24 non-CG advance

Cycle 59's Rome bass profile v1 is the campaign's first M-V4-PROFILES-1 cell filled since the range that closed Chicken Grease at c24:

- Path: data/v4/profiles/51e433ade2a845e1/bass.json
- SHA: a4d4b47ebd9edd0e...
- Top-1: bank 0, program 4, composite 323.55
- Distance: emb_cos_vggish = 0.1677 (well under the 0.40 degenerate-floor upper bound)
- Verdict: SF2_CONFIRMED_provisional (new unfrozen enum label)
- supersedes_path: string pointing at the c24 SF2_RULED_OUT Rome bass verdict per c14 lemma
- Replay proof: REPLAY_PROOF_HOLDS, replay SHA 594f7d4b... under byte-determinism ×2

The landing demonstrates that the operator-omnibus pivot is workable end-to-end (research → sweep → fine-fit → landing → replay-proof) for a single cell without preservation-spin drag. It also demonstrates that OP-1 serial-lock engagement + release on the extended fine_fit_sf2_v2.py driver works under the current disk margin.

Six operator escalation memos formally closed on the substantive side

All six previously-standing operator escalations are now substantively resolved:

Escalation	Resolution
M-V4-METRIC-SEMANTICS-c16	Operator 2026-09-04 distance-semantics ruling
SHOWCASE-1-non-cg-bass-acceptance-policy	Operator 2026-09-05 omnibus point 3 (OPT1 extension)
CERT-composite-fp-drift-adjudication-c32	Operator 2026-09-05 omnibus (Path A implicit via campaign-wide lift)
CERT-fine-fit-sf2-drums-legacy-halt	Subsumed into composite-FP-drift resolution
CERT-fine-fit-sf2-v2-legacy-halt	Subsumed into composite-FP-drift resolution
CERT-fine-fit-sf2-guitar-legacy-halt	Subsumed into composite-FP-drift resolution

None of the six memos was resurrected this range. All are closed on the substantive side; the range's audit correctly notes there were no preservation-spin sidecars emitted for them per the operator point-4 directive.

Read-only anchors held; discipline invariants met

docs/emitter_exemption_policy.md (SHA fd2c33a7...) unchanged this range — no code touched in the emitter-writer-boundary. long_exposure/ ABSENT throughout; the c34 OPT_B exemption chain remains intact (no chain-supersede emitted because the cadence exited). Canonical 7-key env_pin_sha256=2ac444c3...a922ca at pre-launch unchanged. M-V4-CERT-1 E2E_DETERMINISM_HOLDS pre-launch; the certificate remained valid across the fine-fit. SF2_CONFIRMED is now formally allowed campaign-wide under the OPT1-extension acceptance policy (which itself is

the operator-authorized lift of the prior absolute prohibition); the new provisional label SF2_CONFIRMED_provisional encodes “acceptance pending sibling-cell replication” honestly.

Test extensions and carryover

Cycle 58 landed two new tests (test_43, test_44) on tests/test_c30_legacy_mode_regression.py extending the c30 legacy-mode regression suite. Cycle 59 did not re-run these tests as no code was touched in the emitter-writer-boundary. Standalone OP-1 serial-lock suite continues to guard the fine-fit invocation path. One small carryover from Cycle 58 remains: I-2 test_12 SHA drift, a mechanical pin update pending. The range’s audit reclassifies this forward to the next cycle with a note that it should not survive another cycle without either being fixed or explicitly deferred with a reason.

Audit outcome

VALIDATED with observations. Zero CRITICAL, zero HIGH, zero MODERATE. Three low-severity observations queued for the next cycle:

- **O-1: SF2_CONFIRMED_provisional is an unfrozen enum label.** The c23 frozen verdict enum does not carry _provisional. The Rome bass verdict uses this new label to encode the OPT1-lifted acceptance-pending-sibling-cell-replication semantics — honest labeling, but the enum extension needs formalization before the next cycle lands additional cells using it. Recommendation: draft a one-line enum-extension addendum to docs/agent_picks_selection_invariants.md (or the verdict registry) enumerating {SF2_CONFIRMED_provisional, SF2_CONFIRMED} semantics and the promotion criterion. Do not retro-relabel the Rome bass verdict; carry the provisional label forward until the promotion criterion is met.
- **O-2: Sweep hygiene — in-flight versus post-pin semantics.** --score-and-delete-per-candidate was passed to the stage-2 fine driver but 217 WAVs accumulated pre-leaderboard-emit; the post-pin unlink loop cleaned up cleanly. The c27 module’s stated contract is per-candidate delete (working audio ≤500 MB); the observed behavior suggests either (a) the flag is not wired end-to-end in stage-2, or (b) delete is deferred until leaderboard fixation. Disk budget was not violated because post-pin cleanup ran, but the c27 module’s stated invariant is stricter than what stage-2 actually enforces. Recommendation: next cycle brief should ask the worker to grep fine_fit_sf2_v2.py for the flag’s read path and confirm the contract; if the flag is decorative in stage-2, either wire it or amend the c27 module docstring to scope the invariant to stage-1 coarse only.
- **O-3: Disk margin thin at close.** 85%-used, 5.9G-free at range close. Above the c27 df-guards’ hard floor but below the comfortable working margin for the next range’s execute-order (Peach Dream + Disco A bass stage-2 + WIG + Disco A drums stage-1/2). Recommendation: next cycle P0.1 disk-clearance step should target ≥15G free before launching the first stage-2 sweep. The Cycle 58 blocker showed the mv → tools/stale/ + explicit _prune_stale_sweep_audio invocation from the c27 module is the correct workaround pattern when rm -rf is refused; next cycle worker should attempt those before declaring blocked.

The audit explicitly frames the range’s shape: the Rome bass landing “is the first substantive post-c24 non-CG bass advance; correctly executes the OPT1 extension; demonstrates the operator-omnibus pivot is workable end-to-end ... for a single cell without preservation-spin drag.” And equally explicitly what it isn’t: “a closure of M-V4-PROFILES — four more bass/drums cells plus guitar/piano/other/vocals stems plus A/B deliveries plus gen batch plus completion report v3 remain per operator directive 5(a)-(g).”

Discussion

Three things about this range are worth naming.

First, the range is the campaign's exit from a fourteen-cycle preservation-only equilibrium into substantive execution under fresh operator authority. The prior range's audit had projected that the preservation-only cadence could continue mechanically until operator adjudication changed the state; both the campaign's discipline (preserve state; do not manufacture scope) and the operator's response (arrive with two coordinated directives that resolve multiple escalations at once) demonstrate that the equilibrium worked as designed. The distance-semantics ruling closes the metric-semantics escalation open since c16 with a formal answer. The omnibus lifts SF2_CONFIRMED campaign-wide under a named acceptance policy, formally retires the preservation cadence, and enumerates the remaining punch list. Each of the six previously-standing escalations closes on the substantive side without the campaign having to re-litigate any of them. The Rome bass landing then demonstrates the pivot works end-to-end — the same driver (extended `fine_fit_sf2_v2.py`) that had been HALTed on floating-point drift under the strict-equality bar is now productive under the operator-authorized acceptance policy.

Second, the SF2_CONFIRMED_provisional enum label is the correct discipline response to a fresh operator authorization that opened a new semantic before naming its terminal criterion. The OPT1 extension lifts SF2_CONFIRMED under best-of-search-across-families but leaves the sibling-cell replication criterion for “provisional → CONFIRMED” promotion unstated. Rather than either (a) emitting SF2_CONFIRMED on the Rome bass profile and quietly conflating single-cell acceptance with sibling-replicated acceptance, or (b) refusing to emit a verdict at all and re-litigating the operator authorization, the Cycle 59 worker introduced an honest provisional label with the exact semantic the pivot opened. The audit's O-1 recommendation is the natural follow-up: formalize the enum extension in `docs/agent_picks_selection_invariants.md` before the next cycle lands additional cells using the label, and carry the Rome bass provisional label forward until the promotion criterion is met. This is the discipline pattern for provisional acceptance under an operator authorization whose full shape has not been finalized — surface it in the label, formalize the label in the invariants doc next cycle, promote per the criterion once it lands.

Third, the sweep-hygiene observation O-2 is a small but real contract-vs-implementation drift that the range's fresh substantive work surfaced. The c27 hygiene module's per-candidate `render→score→delete` contract was the operational-hygiene backbone the fourteen-cycle preservation cadence had been maintaining as read-only through six-driver legacy-mode regression proofs. When the first substantive stage-2 fine fit actually ran under the module, the observed behavior deviated: the audio accumulated pre-leaderboard-fixation and was cleaned up post-pin rather than per-candidate. The disk budget was not violated, so the observed behavior is defensible — but the c27 module's stated invariant is stricter than what stage-2 actually enforces. This is the kind of gap that only substantive execution surfaces; it was invisible during the preservation-only cadence because no fine fit was running. The audit's recommendation to grep the flag's read path in `fine_fit_sf2_v2.py` and either wire it or scope the module docstring is the correct next-cycle cleanup. The larger pattern: exiting a preservation-only cadence exposes contract drift that only-substantive-work reveals, and the first substantive-work cycles after such an exit should expect and honestly log such gaps rather than declare victory prematurely.

Open questions

- **SF2_CONFIRMED_provisional enum-extension formalization (O-1).** Draft a one-line addendum to `docs/agent_picks_selection_invariants.md` (or the verdict registry) enumerating

{SF2_CONFIRMED_provisional, SF2_CONFIRMED} semantics and the promotion criterion (sibling-cell replication being the operator-omnibus-implied criterion; the exact shape needs operator confirmation or explicit deference). Carry the Rome bass provisional label forward until the criterion is met.

- **Sweep-hygiene per-candidate-delete contract clarity (O-2).** Grep `fine_fit_sf2_v2.py` for the `--score-and-delete-per-candidate` flag's read path; if decorative in stage-2, either wire it end-to-end or amend the `c27` module docstring to scope the invariant to stage-1 coarse only.
- **Disk margin management (O-3).** Next cycle's P0.1 disk-clearance step should target $\geq 15\text{G}$ free before launching the first stage-2 sweep. Use the `mv → tools/stale/ + explicit _prune_stale_sweep_audio` workaround pattern established at Cycle 58 when `rm -rf` is refused, before declaring blocked.
- **Carryover I-2 (test_12 SHA drift).** Mechanical pin update pending since Cycle 58. Should not survive another cycle without either being fixed or explicitly deferred with a stated reason.
- **Remaining M-V4-PROFILES-1 punch list per operator omnibus 5(a)-(g).** Four more bass/drums cells: Peach Dream bass stage-2 (below-floor at 0.4437 in the `c23` stage-1; note this exceeds the 0.40 degenerate-floor upper bound under the formalized distance semantics — needs a distance-under-lifted-acceptance re-check); Disco A bass stage-2 (0.2443, below-floor); WIG bass stage-2 (0.3055, below-floor); WIG + Disco A drums stage-1/2. Plus guitar/piano/other/vocals stems, A/B deliveries per song, generator batch, and completion report `v3`.
- **Rome bass sibling-cell replication.** If sibling-cell replication is the operator-implied `SF2_CONFIRMED` promotion criterion, then the Rome bass provisional acceptance promotes only when at least one additional song's bass cell lands under the same OPT1 extension. Cycle 60 or later would be the earliest possible promotion.
- **M-V4-EAR-1 / M-V4-RULES-1 / M-V4-GEN-1 / M-V4-CLOSE-1 status.** Unchanged this range. `M-V4-RULES-1` scaffold landed `c20`; substantive implementation queued. `M-V4-EAR-1` not yet opened. `M-V4-GEN-1` conditional on `M-V4-RULES` + `M-V4-EAR`. `M-V4-CLOSE-1` `c24` amendment landed; a completion report `v3` is in the operator punch list.
- **Preservation-only cadence formally retired.** Per operator omnibus point 4. Future cycles do not emit per-cycle preservation sidecars for the Priority 0 escalation set, the Priority 1 emitter-writer-boundary chain, the Priority 2 POR shadow-drift stand-pat chain, the Priority 5 Track B/C/D deferral rollup, or the Priority 8 consolidation-proposal HOLD. These surfaces are now maintained by substantive advance rather than by heartbeat preservation.

Appendix: Provenance

Directive. Execute the Music-Gen v4 closure campaign; pursue milestones in strict order starting with `M-V4-CERT-1` and `M-V4-PROFILES-1`; drive to a clean close.

Cycle range. cycles 57–59.

Working directory. `/home/user/long-exposure-runs/music-gen`.

Session references.

- Cycle 57 researcher `a71a8a21-5fc5-4512-9ae2-fae314af720c`; worker `5ac5afeb-d9e4-446d-a1aa-c2c494b5c726`; auditor `008ce8cf-8ca1-437e-9414-ddaed7b04c9b`.
- Cycle 58 researcher `f41e31a8-7660-43ef-93e3-5518a646dd41`; worker `0113c8fa-7b3a-4534-ada6-2c0049b46427`; auditor `0fddeb6a-0c72-4b69-a2d5-cf1c5468c256`.
- Cycle 59 researcher `9fe34f5f-bec8-418a-bea3-0d88a8f9ebff`; worker `d1868226-3f22-4be7-810e-`

3b0768c83bef; auditor b4207f1a-a932-4c6d-b292-ed060ac41eb.

Audit verdict. VALIDATED with observations. Zero CRITICAL, zero HIGH, zero MODERATE. Three low-severity observations queued for the next cycle (O-1 SF2_CONFIRMED_provisional enum-extension formalization; O-2 sweep-hygiene per-candidate-delete contract clarity in fine_fit_sf2_v2.py; O-3 disk margin thin at 85% used / 5.9G free at range close). Carryover I-2 (test_12 SHA drift) reclassified forward.

Operator directives arrived this range.

- **Operator 2026-09-04 distance-semantics ruling.** embedding_cos_vggish is distance; 0.40 is degenerate-floor upper bound; candidates below 0.40 eligible for acceptance under acceptance policy; candidates above 0.40 degenerate.
- **Operator 2026-09-05 omnibus.** Point 3: SF2_CONFIRMED lifted campaign-wide under OPT1-extension best-of-search-across-families acceptance policy. Point 4: preservation-spin cadence formally retired; no per-cycle preservation sidecars from this point. Points 5(a)-(g): remaining M-V4-PROFILES-1 punch list — four more bass/drums cells + guitar/piano/other/vocals stems + A/B deliveries + generator batch + completion report v3.

Terminal deliverables landed this range.

- Formal replanning under the two operator directives (Cycle 57).
- P0.1 disk-clearance workaround pattern established (mv → tools/stale/ + explicit_prune_stale_sweep_audio invocation from c27 module when rm -rf is refused) (Cycle 58).
- Two new tests (test_43, test_44) on tests/test_c30_legacy_mode_regression.py extending the c30 legacy-mode regression suite (Cycle 58).
- data/v4/profiles/51e433ade2a845e1/bass.json (SHA a4d4b47ebd9edd0e...) — Rome bass profile v1, top-1 bank 0 program 4, composite 323.55, emb_cos_vggish distance 0.1677, verdict SF2_CONFIRMED_provisional, supersedes_path string → c24 SF2_RULED_OUT verdict per c14 lemma (Cycle 59).
- bass.replay_proof.json — REPLAY_PROOF_HOLDS, replay SHA 594f7d4b... under byte-determinism ×2 (Cycle 59).
- One milestone-level M-V4-PROFILES ledger event honoring the v4-relaxed cadence (Cycle 59).
- OP-1 serial-lock engaged and released cleanly on fine_fit_sf2_v2.py invocation (Cycle 59).

Six operator escalations formally closed on the substantive side this range.

- M-V4-METRIC-SEMANTICS-c16 — closed by operator 2026-09-04 distance-semantics ruling.
- SHOWCASE-1-non-cg-bass-acceptance-policy — closed by operator 2026-09-05 omnibus point 3.
- CERT-composite-fp-drift-adjudication-c32 — closed by operator 2026-09-05 omnibus (Path A implicit via campaign-wide lift).
- CERT-fine-fit-sf2-drums-legacy-halt, CERT-fine-fit-sf2-v2-legacy-halt, CERT-fine-fit-sf2-guitar-legacy-halt — subsumed into composite-FP-drift resolution.

None of the six memos resurrected. No preservation-spin sidecars emitted per operator point-4 directive.

Preservation surfaces at range close.

- Priority 1 (emitter-writer-boundary): docs/emitter_exemption_policy.md (SHA fd2c33a7...) unchanged; long_exposure/ still ABSENT; chain-supersede cadence formally retired per operator point 4. c34 OPT_B exemption remains valid but no longer requires per-cycle re-attestation.

- Priority 2 (POR shadow-drift stand-pat): cadence formally retired per operator point 4.
- Priority 0, 5, 8 preservation sidecars: cadence formally retired per operator point 4.

M-V4-PROFILES cell tally at range close.

- Chicken Grease: 5/5 terminal (validated at c24).
- Rome bass: SF2_CONFIRMED_provisional (new; pending sibling-cell replication for promotion).
- Rome drums, guitar (family-2 stem-sampled per c15 pattern), piano, other, vocals: unfilled.
- WIG bass: STILL_INDETERMINATE at 0.3055 below-floor; queued for next range under OPT1 extension.
- WIG drums, guitar (empty per MIDI-probe → NULL), piano, other, vocals: unfilled.
- Disco A bass: STILL_INDETERMINATE at 0.2443 below-floor; queued for next range under OPT1 extension.
- Disco A drums, guitar, piano, other, vocals: unfilled.
- Peach Dream bass: SF2_RULED_OUT at 0.4437 (above the 0.40 formalized upper bound under distance semantics — needs a distance-under-lifted-acceptance re-check as it exceeds the degenerate floor).
- Peach Dream drums, guitar (empty per MIDI-probe → NULL), piano, other, vocals: unfilled.

Environment pin. Canonical 7-key env_pin_sha256=2ac44c36298d6ada0579aba1a9160a5881703a4e628f5ccdd828b842 at pre-launch unchanged this range; FD-16(a) re-issue not triggered. M-V4-CERT-1 E2E_DETERMINISM_HOLDS pre-launch; certificate remained valid across the fine-fit. FD-16(c) replay-proof ×2 per render family per song satisfied on the Rome bass profile.

Discipline guards asserted (AST-scannable). No PRNG imports, no sidecar_nonfactor, no VST3 state APIs, no --verify-det bypass, /usr/bin/python3 interpreter guard on the fine-fit invocation. supersedes_path typed as string per c14 lemma on the Rome bass profile (never list). OP-1 serial-lock engaged and released cleanly. FD-1 honored: no tuning, no retry, no fallback observed in the stage-2 fine fit. No wait-on-operator memo (banned per operator 2026-09-03 point 2). None of the six previously-standing operator escalation memos resurrected.

Ledger routing. Single milestone-level ledger event under M-V4-PROFILES emitted at Cycle 59 (v4-relaxed cadence). No orphaned promises opened; the Rome bass promise closed by the profile + replay-proof pair as expected.

Milestone status at range close.

- M-V4-CERT-1 — validated (E2E_DETERMINISM_HOLDS on the v3 spine; pre-launch verification passed each cycle).
- M-V4-PROFILES-1 CG (5/5 instruments) — validated (unchanged).
- M-V4-PROFILES-1 non-CG bass — Rome SF2_CONFIRMED_provisional (new this range); WIG STILL_INDETERMINATE (queued); Disco A STILL_INDETERMINATE (queued); Peach Dream SF2_RULED_OUT (above 0.40 upper bound under formalized distance semantics; needs re-check).
- M-V4-PROFILES-1 non-CG drums — 0/4 (queued per operator punch list).
- M-V4-PROFILES-1 non-CG guitar — 0/2 substantive (WIG + Peach Dream guitar are NULL by earlier MIDI-probe; Rome + Disco A guitar queued).
- M-V4-PROFILES-1 piano / other / vocals stems — queued per operator punch list.
- M-V4-SHOWCASE-1 CG — LANDS_pending_operator (cg_ab_mix.wav SHA 6e13e007... byte-identical since c17).
- M-V4-SHOWCASE-1 non-CG — unblocked at the policy level by operator omnibus; A/B deliveries queued per punch list.

- M-V4-RULES-1 — scaffold landed c20; substantive implementation queued.
- M-V4-EAR-1 — not yet opened.
- M-V4-GEN-1 — conditional on M-V4-RULES + M-V4-EAR.
- M-V4-CLOSE-1 — c24 amendment landed; completion report v3 queued per operator punch list.

Next-cycle first tasks. (i) P0.1 disk-clearance step targeting $\geq 15\text{G}$ free using the `mv -> tools/stale/ + explicit _prune_stale_sweep_audio` pattern before launching the first stage-2 sweep. (ii) O-1: draft one-line enum-extension addendum to `docs/agent_picks_selection_invariants.md` formalizing `{SF2_CONFIRMED_provisional, SF2_CONFIRMED}` semantics and the promotion criterion; do not retro-relabel Rome bass. (iii) O-2: `grep fine_fit_sf2_v2.py` for `--score-and-delete-per-candidate` read path; either wire end-to-end or amend c27 module docstring to scope invariant to stage-1 coarse only. (iv) I-2 mechanical fix: update test_12 SHA drift pin or explicitly defer with stated reason. (v) Execute operator punch list 5(a)-(g): Peach Dream bass re-check under distance-under-lifted-acceptance semantics; WIG + Disco A bass stage-2; WIG + Disco A drums stage-1/2; guitar/piano/other/vocals stems per applicable songs; A/B deliveries; generator batch; completion report v3. Operator ear remains LANDS authority post-hoc per FD-6.